HPV Vaccination Simulation Model

Cost-effectiveness of extending vaccination age cap within VHA

2026-05-26

Model overview

Cohort

- 1.5M male veterans, ages 26–45 at entry
- Male only (VHA OPC burden is overwhelmingly male)
- 75-year horizon, annual timesteps
- 3% discount rate (costs and QALYs)
- Common random numbers across scenarios

5 health states

Healthy → Infected → Persistent → Cancer → Dead
with background mortality from all states (SSA 2021 male life table)

Agent attributes: age, smoker, heavy alcohol use, vaccination status, health state, infection and cancer duration

5 scenarios — vaccination age cap

Scenario	Age cap
Status quo	26
Expansion 1	30
Expansion 2	35
Expansion 3	40
Expansion 4	45 (FDA-approved)

Outcome measures: discounted QALYs, vaccination costs, OPC treatment costs, OPC cases averted, OPC deaths averted, ICER

State transitions

Transition	Mechanism	Parameters
Healthy $\rightarrow$ Infected	Annual probability	$p = 0.041/$yr; reduced 90% if vaccinated; increased by alcohol (RR 1.43), smoking (RR 1.15)
Infected $\rightarrow$ Healthy	Annual clearance	0.60 (yr 0–1), 0.40 (yr 2+); reduced by smoking (RR 0.70)
Infected $\rightarrow$ Persistent	One-shot roll at $ $2 yr	$p = \mathbf{0.020}$ *(calibrated)*; reduced 85% if vaccinated
Persistent $\rightarrow$ Cancer	Weibull latency	shape $= 3$, scale $= 20$ (mean ≈ 18 yr); accelerated by smoking/alcohol
Cancer $\rightarrow$ Dead (OPC)	Competing risks	$p_{\text{OPC}} =$ excess rate$/($background $+$ excess); $\approx 28\%$ attributed

Key parameters

Parameter	Value	Source
Annual HPV acquisition	0.041/yr	Dube Mandishora 2024 (HIM Study US, $n = 834$)
Clearance probability	0.60 (yr 0–1) / 0.40 (yr 2+)	Kreimer 2013 (HIM cohort)
Persistence probability	**0.020 (calibrated)**	**Target: 6–9/100k PY (Saxena 2022 VA); achieved 6.59/100k PY**
OPC latency (Weibull)	shape $= 3$, scale $= 20$, mean ≈ 18 yr	Consistent with reviewer: "likely 10–30 years"
OPC 5-year survival	$\approx 79\%$	Ang 2010 (HPV+ OPC, $n = 323$)
Vaccine efficacy — acquisition	90% reduction	Damgacioglu 2022; Chaturvedi 2018
Vaccine efficacy — persistence	85% reduction	Extrapolated (no direct trial data)
Vaccine uptake	15%/yr if eligible	**Placeholder** — Aim 1 will supply age-stratified rates
Vaccine cost	\$550/series	CDC price list, Dec 2024
OPC treatment cost	\$85,000/yr (active phase)	Saxena 2022 VA, CPI-adjusted to 2024 USD
Background mortality	SSA 2021 period life table	SSA Trustees Report 2024

Progress since grant review

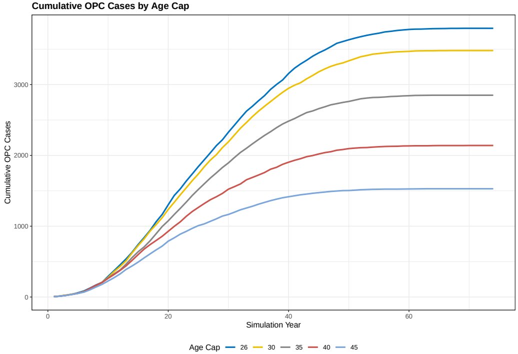
Addressed

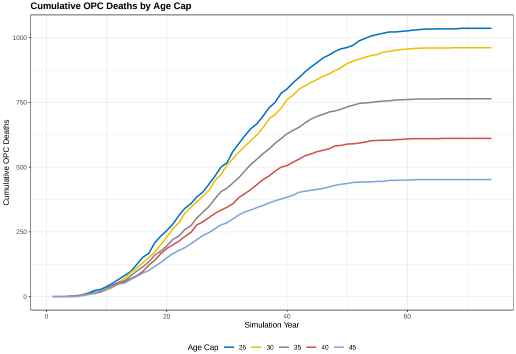
- **Sample size**: $10\text{k} \to 1.5\text{M}$ (policy-relevant VHA population)
- **VA OPC incidence calibration**: $p_{\text{pers}} = 0.020$ yields 6.59/100k PY; target 6–9 (Saxena 2022)
- **OPC latency**: Weibull mean ≈ 18 yr (scale $= 20$), consistent with reviewer range of 10–30 yr
- **OPC death attribution**: competing risks approach ($\approx 28\%$ CFR); flat assignment previously gave 99.6%
- **ICER stability**: common random numbers (shared population + mortality draws); all 5 scenarios on efficient frontier

Remaining

- Vaccination uptake: flat 15% placeholder $\to$ replace with Aim 1 real data (age-stratified by 5-year band)
- Probabilistic sensitivity analysis (PSA) and tornado diagram not yet run
- Veteran mortality multiplier $= 1.0$ (general US male); literature suggests ~ 1.1–1.2 for VA
- No VE discount for ages 40–45 (immunogenicity non-inferior by trial, but real-world effectiveness uncertain)
- MSM removed from model (was in original design; out of current scope)

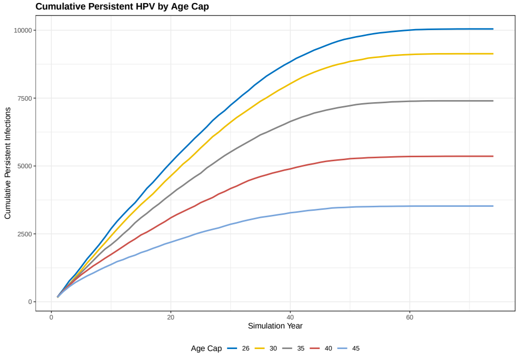
Results: OPC cases and deaths

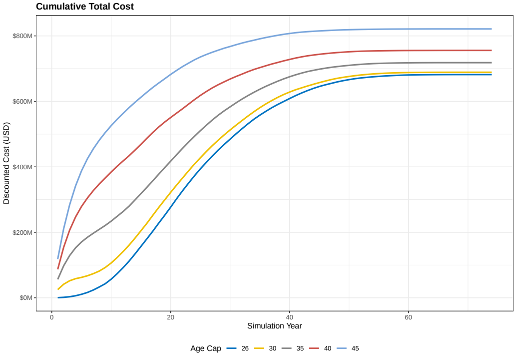




$n = 1{,}500{,}000$ male veterans; CRN; calibrated model ($p_{\text{pers}} = 0.020$, 6.59 HPV+ OPC per 100k PY). Age cap 26 = status quo.

Results: persistent infections and costs





Left: persistent HPV infections (intermediate endpoint). Right: cumulative discounted costs (vaccine + OPC treatment). Higher age caps increase vaccine spending but offset via reduced cancer costs.